

Section : Physics

Q.1 Two identical capacitors are connected in parallel combination. Their equivalent capacitance is C_1 . If the plate area is halved and separation between plates is doubled for both the capacitors, their equivalent capacitance now is C_2 . Find the ratio C_1/C_2 .

1. 16:1
2. 4:1
3. 8:1
4. 1:4

Options 1. 1

2. 2
3. 3
4. 4

Q.2 A current of 10 A is flowing in a wire of length 1.5 m. When placed in a uniform magnetic field of 2 T, it experiences a force 15 N. The angle between the magnetic field and the direction of flow of current is

1. 60°
2. 30°
3. 90°
4. 0°

Options 1. 1

2. 2
3. 3
4. 4

Q.3 A charged particle is released from rest in a region of a uniform magnetic field along the direction of the magnetic field. The path of the charged particle in the magnetic field will be

1. straight line
2. helical
3. circular
4. elliptical

Options 1. 1

2. 2
3. 3
4. 4

Q.4 A plane electromagnetic wave of frequency 50 MHz travels in free space in the X- direction. At a particular point in space and time, the magnetic field is 0.003 G in the Z-direction. The magnitude and direction of the electric field is

1. 9 V/m, along the negative Y-direction
2. 1×10^{-15} V / m along the Y-direction
3. 90 V/m along the Y-direction
4. 1×10^{15} V / m along the Y-direction

Options 1. 1

2. 2
3. 3
4. 4

Q.5 Match List-I with List-II

List-I

List-II

(EM Waves)

(Production)

(A) Infra-red rays

(I) Electrons in atoms emit light when they move from one energy level to a lower level

(B) Visible light rays

(II) Radioactive decay of the nucleus

(C) Ultraviolet rays

(III) Vibration of molecules

(D) Gamma rays

(IV) Inner shell electrons in atoms moving from one energy level to a lower level

Choose the correct answer from the options given below:

1. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.6 The resistivity of a material depends on

1. length, area of cross section and temperature of material
2. length and area of cross section of material
3. nature and temperature of material
4. temperature of material only

Options 1. 1

2. 2
3. 3
4. 4

Q.7 In a Young's double slit experiment, the distance between slits is 2.5 mm and the screen is placed 2 m away from the slits. The fringe width for a light of wavelength 400 nm is

1. 2.5 mm
2. 3.32 mm
3. 0.16 mm
4. 0.32 mm

- Options 1. 1
2. 2
 3. 3
 4. 4

Q.8 A conducting circular loop is placed in a uniform magnetic field, $B = 0.025 \text{ T}$ with its plane perpendicular to the magnetic field. The radius of the loop is made to shrink at a constant rate of 2 mm s^{-1} . The induced emf when the radius is 2 cm, is

1. $\pi \mu\text{V}$
2. $2\pi \mu\text{V}$
3. $4\pi \mu\text{V}$
4. $5\pi \mu\text{V}$

- Options 1. 1
2. 2
 3. 3
 4. 4

Q.9 Identify the incorrect statements from the following.

(A) A charged particle of charge e is moving through a magnetic field, work will be done by the magnetic field on the charge.

(B) A conductor with free electrons is placed in a magnetic field. No current is flowing through the conductor. No magnetic force acts on the conductor.

(C) A charged particle is moving perpendicular to the magnetic field. The magnetic force does not change the speed of the particle but changes its direction.

(D) A charged particle is moving in a straight direction without any change of direction. It shows that no magnetic field is there.

Choose the *correct* answer from the options given below:

1. (A) and (B) only
2. (A) and (C) only
3. (C) and (D) only
4. (A) and (D) only

- Options 1. 1
2. 2
 3. 3
 4. 4

Q.10 In a Young's double-slit experiment, the slits are separated by 0.56 mm and the screen is placed 2.8 m away from the slits. The distance between the central bright fringe and the fourth bright fringe is measured to be 1.2 cm. Determine the wavelength of light used in the experiment.

1. 600 nm
2. 400 nm
3. 800 nm
4. 700 nm

- Options 1. 1
2. 2
 3. 3
 4. 4

Q.11 A solenoid with an iron core and a bulb are connected to an a.c. source. The bulb is glowing with certain brightness. When the iron core is removed from the solenoid, the brightness of the bulb

1. increases
2. decreases
3. remains same
4. increases and later decreases

- Options 1. 1
2. 2
 3. 3
 4. 4

Q.12 A straight long wire carrying a current of 30 A is placed in an external uniform magnetic field of $4 \times 10^{-4} \text{ T}$ parallel to the current. The magnitude of the resultant magnetic field at a point 2 cm away from the wire is

1. $1.5 \times 10^{-4} \text{ T}$
2. $3 \times 10^{-4} \text{ T}$
3. $2 \times 10^{-4} \text{ T}$
4. 10^{-4} T

- Options 1. 1
2. 2
 3. 3

Q.13 An electron revolves around a very long line charge having density $2 \times 10^{-6} \text{ C m}^{-1}$. The kinetic energy of the electron is

1. $5 \times 10^{-17} \text{ J}$
2. $2.88 \times 10^{-16} \text{ J}$
3. $2.88 \times 10^{-15} \text{ J}$
4. $5 \times 10^{-15} \text{ J}$

Options 1. 1

2. 2
3. 3
4. 4

Q.14 For n-type semiconductor, which of the following statements is true?

1. Trivalent atoms are the dopant and electrons are majority carriers.
2. Trivalent atoms are the dopant and holes are majority carriers.
3. Pentavalent atoms are the dopant and electrons are minority carriers.
4. Pentavalent atoms are the dopant and holes are the minority carriers.

Options 1. 1

2. 2
3. 3
4. 4

Q.15 If the wavelength of light in an experiment on photoelectric effect is increased, then

- (A) the photoelectric emissions will not take place
 - (B) the photoelectric emission may or may not take place
 - (C) the stopping potential will increase.
 - (D) the stopping potential will decrease
- Choose the correct answer from the options given below:

1. (B) and (D) only
2. (B) and (C) only
3. (A) and (C) only
4. (A) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.16 The transverse nature of light is shown by

1. Interference of light
2. Refraction of light
3. Polarisation of light
4. Dispersion of light

Options 1. 1

2. 2
3. 3
4. 4

Q.17 α , β and γ rays emitted by a radioactive substance are allowed to pass through a region containing a magnetic field at right angles to their path. The energy gained will be

1. maximum for α rays
2. maximum for β rays
3. maximum for γ rays
4. zero for all of them

Options 1. 1

2. 2
3. 3
4. 4

Q.18 Assume the earth to be a spherical conductor of radius 6400 km. Calculate the capacitance of earth.

1. 711 nF
2. 7.11 nF
3. 7.11 μF
4. 711 μF

Options 1. 1

2. 2
3. 3
4. 4

Q.19 In a photoelectric effect, the number of photo electrons emitted per second is directly proportional to

1. frequency of the incident radiation
2. wavelength of the incident radiation
3. intensity of the incident radiation
4. speed of the incident radiation

Options 1. 1

2. 2
3. 3

Q.20 A radioactive material with half life of 6.25 years undergoes beta decay. How much fraction of material is decayed after 25 years:

1. $1/16$
2. $15/16$
3. $1/8$
4. $7/8$

Options 1. 1

2. 2
3. 3
4. 4

Q.21 The electric field in a copper wire of cross-sectional area 4 mm^2 carrying a current of 1 A and having electrical conductivity of $6.25 \times 10^7 \text{ Sm}^{-1}$ is

1. $4 \times 10^{-3} \text{ Vm}^{-1}$
2. $4 \times 10^{-2} \text{ Vm}^{-1}$
3. $5 \times 10^{-3} \text{ Vm}^{-1}$
4. 0.4 Vm^{-1}

Options 1. 1

2. 2
3. 3
4. 4

Q.22 Monochromatic light of wavelength 600 nm is incident from air on to water surface. The frequency of refracted light is

1. $5 \times 10^5 \text{ Hz}$
2. $5 \times 10^{14} \text{ Hz}$
3. $2 \times 10^{13} \text{ Hz}$
4. $3.75 \times 10^{14} \text{ Hz}$

Options 1. 1

2. 2
3. 3
4. 4

Q.23 A system of three charges $-q$, $+Q$ and $-q$ are placed at three points at equal distances on a straight line in the same order. If the potential energy of the system is found to be zero, find the ratio Q/q .

1. $\frac{1}{2}$
2. $\frac{4}{1}$
3. $\frac{1}{4}$
4. $\frac{2}{1}$

Options 1. 1

2. 2
3. 3
4. 4

Q.24 An alternating voltage $V = V_0 \sin(\omega t + \pi/2)$ is applied across an ac circuit. As a result, a current $I = I_0 \sin(\omega t + \pi/6)$ flows in it. The power factor of the circuit is

1. zero
2. 0.5
3. 1
4. $\frac{\sqrt{3}}{2}$

Options 1. 1

2. 2
3. 3
4. 4

Q.25 The expression of the magnetic energy stored in a solenoid in terms of magnetic field B , cross-sectional area A and length l of the solenoid, is:

1. $U_B = \frac{1}{2\mu_0} B^2 Al$
2. $U_B = \frac{1}{\mu_0} B^2 Al$
3. $U_B = \frac{1}{2\mu_0} B^2 \frac{A}{l}$

$$4. U_B = \mu_0 B^2 A l$$

Options 1. 1

2. 2
3. 3
4. 4

Q.26 Which of the following statements are correct?

- (A) In reverse bias of a p-n junction, negative terminal of the battery is connected to p-side of the junction,
 (B) Reverse bias does not support the potential barrier at the junction.
 (C) In reverse bias, a few minority charge carriers cross the junction.
 (D) Resistance of p-n junction is high in reverse bias, reducing the current to a negligible value.

Choose the correct answer from the given options.

1. (A) and (D) only
2. (A), (C) and (D) only
3. (A) and (C) only
4. (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.27 Match List-I with List-II

List-I

Optical instrument

- (A) Magnifying power m of a simple microscope
 (B) Magnifying power m of a simple microscope if the image is at infinity
 (C) Magnifying power m of a compound microscope
 (D) Magnifying power m of a telescope

List-II

Magnifying power

- (I) $m = 1 + (D/f)$
 (II) $m = (L/f_o) \times (D/f_e)$
 (III) $m = f_o / f_e$
 (IV) $m = D/f$

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (IV), (C) - (II), (D) - (III)
2. (A) - (I), (B) - (IV), (C) - (III), (D) - (II)
3. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)
4. (A) - (IV), (B) - (I), (C) - (III), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.28 A wire of length 10 m and diameter 0.4 mm has a resistance of 2 Ω . The resistivity of the wire is

1. $2.5 \times 10^{-8} \Omega m$
2. $1.5 \times 10^{-8} \Omega m$
3. $1 \times 10^{-7} \Omega m$
4. $2 \times 10^{-6} \Omega m$

Options 1. 1

2. 2
3. 3
4. 4

Q.29 The light rays travel from a glass slab with refractive index μ_1 to the second slab with refractive index μ_2 .

The refractive index of the second slab with respect to the first one is:

(Assume that the two slabs have the same thickness)

1. $\mu_2 = \mu_1 / \mu_2$
2. $\mu_2 = \mu_2 / \mu_1$
3. $\mu_2 = \mu_1 \times \mu_2$
4. $\mu_1 = \mu_1 / \mu_2$

Options 1. 1

2. 2
3. 3
4. 4

Q.30 Resonant frequency of a LCR series circuit is 16 kHz. If the capacitance is increased to 16 times, then the new resonant frequency will become

1. 16 kHz
2. 32 kHz
3. 8 kHz
4. 4 kHz

Options 1. 1

2. 2

3. 3

4. 4

Q.31 The ratio of the wavelengths associated with electrons accelerated with the potentials of 64 V and 81 V, respectively is

1. 1:1
2. 9:4
3. 9:8
4. 8:9

Options 1. 1

2. 2
3. 3
4. 4

Q.32 10^{18} electrons pass from point A to point B in 0.1 s in a conductor. The value of the current is

1. 2.6 A
2. 0.6 A
3. 16 A
4. 1.6 A

Options 1. 1

2. 2
3. 3
4. 4

Q.33 In a p-n junction, the magnitude of the drift current is lower than the magnitude of the diffusion current when the junction is

1. either forward biased or reverse biased
2. forward biased
3. reverse biased
4. under equilibrium

Options 1. 1

2. 2
3. 3
4. 4

Q.34 A current carrying solenoid of cross-sectional area 2 cm^2 and 2000 turns has a magnetic moment of 0.80 Am^2 . The current flowing through the solenoid is

1. 0.8 A
2. 2 A
3. 4 A
4. 0.2 A

Options 1. 1

2. 2
3. 3
4. 4

Q.35 The electric potential at a point P due to a point charge of $9 \mu\text{C}$ located 10 cm away from point P is

1. $8.1 \times 10^6 \text{ V}$
2. $8.1 \times 10^7 \text{ V}$
3. $8.1 \times 10^4 \text{ V}$
4. $8.1 \times 10^5 \text{ V}$

Options 1. 1

2. 2
3. 3
4. 4

Q.36 A small telescope has an objective lens of focal length 100 cm and an eye-piece of focal length 5 cm. The magnifying power of the telescope when the final image is formed at the least distance of distinct vision will be

1. 24
2. 20
3. 105
4. 30

Options 1. 1

2. 2
3. 3
4. 4

Q.37 Two wires A and B of the same material and same mass have radii $2r$ and r , respectively. If the resistance of wire A is 4Ω , then resistance of B will be

1. 64Ω
2. 4Ω
3. 32Ω
4. 16Ω

Options 1. 1

2. 2
3. 3
4. 4

Q.38 A battery of emf 10 V and internal resistance $3\ \Omega$ is connected to an external resistor. If the current in the circuit is 0.5 A, then the voltage drop across the external resistor will be

1. 8.5 V
2. 8 V
3. 17 V
4. 11.5 V

Options 1. 1

2. 2
3. 3
4. 4

Q.39 Two capacitors of capacitances of $6\ \mu\text{F}$ and $12\ \mu\text{F}$ are connected in series with a 12 V battery. The total energy stored in the combination is

1. 1296 μJ
2. 96 μJ
3. 192 μJ
4. 288 μJ

Options 1. 1

2. 2
3. 3
4. 4

Q.40 If a wave gets refracted into a denser medium, which of the following is true?

1. frequency, wavelength and speed increase
2. frequency, wavelength and speed decrease
3. frequency remains constant but wavelength decreases and speed increases
4. frequency remains the same but both wavelength and speed decrease

Options 1. 1

2. 2
3. 3
4. 4

Q.41 The resistance of a wire is $R\ \Omega$. What will be its new resistance R_{new} if it is stretched to n times its original length?

1. $R_{\text{new}} = nR$
2. $R_{\text{new}} = n^2 R$
3. $R_{\text{new}} = R$
4. $R_{\text{new}} = \frac{R}{n^2}$

Options 1. 1

2. 2
3. 3
4. 4

Q.42 The half life of a radioactive substance is 20 s. The decay constant is

1. 0.0346s^{-1}
2. 0.693s^{-1}
3. $2 \times 10^{-6}\text{s}^{-1}$
4. $3.46 \times 10^{-6}\text{s}^{-1}$

Options 1. 1

2. 2
3. 3
4. 4

Q.43 A square of side L meters lies in the xy -plane in a region where the magnetic field is given by $\mathbf{B} = B_0(2\mathbf{i} + 4\mathbf{j} + 5\mathbf{k})\ \text{T}$, where B_0 is constant. The magnitude of flux passing through the square is

1. $5B_0L^2\ \text{Wb}$
2. $4B_0L^2\ \text{Wb}$
3. $2B_0L^2\ \text{Wb}$
4. $B_0L^2\ \text{Wb}$

Options 1. 1

2. 2
3. 3
4. 4

Q.44 Match List-I with List-II

List-I

Parameter of Magnetic Dipole

(A) Magnetic moment

(B) Magnetic field on the equatorial line (II) $m \times B$

(C) Magnetic field on axial line (III) IA

(D) External torque in magnetic field

List-II

Formula

(I) $\mu_0 2m / (4\pi x^3)$

(III) IA

Extra close brace or missing open brace

Choose the correct answer from the options given below:

1. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)
2. (A) - (II), (B) - (III), (C) - (I), (D) - (IV)
3. (A) - (I), (B) - (IV), (C) - (III), (D) - (II)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.45 Arrange the following in the decreasing order of their wavelengths.

- (A) microwaves
(B) ultraviolet rays
(C) infrared rays
(D) gamma rays

Choose the correct answer from the options given below:

1. (A) (C), (B), (D)
2. (A), (D), (B), (C)
3. (D), (B), (C), (A)
4. (A), (B), (C), (D)

Options 1. 1

2. 2

3. 3

4. 4

Q.46 A tightly wound 100 turn coil of radius 5 cm is carrying a certain amount of current. The magnitude of the magnetic field at the center of the coil is $2\pi \times 10^{-4}$ T. The value of current flowing through the coil is

1. 0.05 A
2. 0.1 A
3. 2 A
4. 0.5 A

Options 1. 1

2. 2

3. 3

4. 4

Q.47 A hollow metal sphere of radius 10 cm is uniformly charged such that the electric potential at a distance of 20 cm from the center is 10 V. The electric potential at a distance of 5 cm from the center of the sphere will be

1. Zero
2. 20 V
3. 5 V
4. 40 V

Options 1. 1

2. 2

3. 3

4. 4

Q.48 Express 16 mg mass into equivalent energy in eV

1. 9×10^{30} eV
2. 9×10^{29} eV
3. 9×10^{16} eV
4. 10^{28} eV

Options 1. 1

2. 2

3. 3

4. 4

Q.49 The electric potential energy for a system of two point charges of 6 μC and -3 μC placed at points x = -9 cm and x = 9 cm, respectively is

1. -0.9 J
2. 9 J
3. -1.8 J
4. 0.3 J

Options 1. 1

2. 2

3. 3

4. 4

Q.50 A railway track running north-south has two parallel rails 1 m apart. Calculate the value of induced emf between the rails, when a train passes at a speed of 25 m/s. The vertical component of earth's magnetic field at that place is $0.3 \times 10^{-4} \text{ Wbm}^{-2}$

1. $1.3 \times 10^{-3} \text{ V}$

2. $1.3 \times 10^3 \text{ V}$

3. $7.5 \times 10^{-4} \text{ V}$

4. 0.13 V

Options 1. 1

2. 2

3. 3

4. 4